Sustainable Heat & Power Generation Technologies

Built Environment

May, 2026

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Sustainable Heat & Power Generation Technologies (A30605)

Short Title: Sust Heat & Power Gen Tech
Faculty Engineering
Department Built Environment
Cluster Engineering
Author DWILLIAMS
Credits 5 **Level:** Intermediate

Description of Module / Aims

The aim of this module is to provide a specialised knowledge of sustainable heat and power generation.

Programmes

	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	2 / 3 / M
ENGR-0044	BEng (Hons) in Sustainable Civil Engineering (WD_ESCIV_B)	3 / 5 / E
ENGR-0044	BEng (Hons) in Sustainable Energy Engineering (WD_ESERG_B)	2 / 3 / M
	BEng (Hons) (WD_ETRIC_B)	2 / 3 / M
ENGR-0094	BEng in Electrical Engineering (WD_EELCT_D)	3 / 5 / M
ENGR-0044	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 7 / M

Indicative Content

- Wind energy
- Hydro energy
- Solar energy
- Energy storage technology and fuel cells
- Waste heat recovery and utilisation
- CHP and trigeneration
- Nuclear power
- Biomass

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the principles of heat and power generation.
2. Analyse the characteristics and performance of alternative energy systems.
3. Apply standard calculation techniques to solve problems involving energy supply and demand.
4. Describe, compare and contrast energy storage technologies.
5. Calculate and compare costs and benefits of different low and zero carbon technologies.

Learning and Teaching Methods

- Lectures
- Presentations
- Interactive Practicals
- Reading
- Exercises
- Discussion

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
Assignment	30%	3

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	48	
Independent Learning	87	

Essential Material

- BSRIA. _Guides and Technical Notes_ BSRIA. ..
- Boyle, G. _Energy Systems & Sustainability_. UK: Oxford University Press, 2003.
- Boyle, G. _Renewable energy - power for a sustainable future_. 2nd ed.. UK: Oxford University Press, 2004.
- CIBSE. _Guides and Technical Notes_ CIBSE. ..
- Manwell, J.F. _Wind Energy Explained_. 2nd ed.. UK: Wiley, 2009.
- O'Hayre, R. _Fuel Cell Fundamentals_. 2nd ed.. UK: Wiley, 2009.
- Rosa, A.V.D. _Fundamentals of Renewable Energy Processes_. UK: Elsevier, 2005.

Requested Resources

- Lecture Room: Loose Seated